

COASTAL CAROLINA, NC, USA

WMO: 723095

Lat: 35.068N

Lon: 77.048W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	26.1	5.9	7.4	28.3	10.6	9.4	32.8	22.9	53.2	20.2	46.8	5.4	350	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	93.3	78.1	91.1	77.2	89.2	76.4	80.3	89.1	79.2	87.4	78.2	85.8	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	145.6	84.0	77.0	140.5	83.1	76.1	136.2	82.1	43.7	89.3	42.6	87.3	41.5	86.0	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.9	15.2	DB	16.6	97.4	3.5	2.1	14.0	98.9	12.0	100.2	10.0	101.3	7.4	102.9	(4)
(5)			WB	15.0	82.9	3.4	1.5	12.5	83.9	10.6	84.8	8.7	85.6	6.2	86.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	44.4	47.4	53.3	62.4	70.0	77.6	80.2	79.2	74.4	64.7	54.5	48.3		
	DBStd	14.73	10.74	9.23	9.08	7.85	6.49	4.51	3.60	3.55	4.80	7.68	8.25	9.81		
	HDD50	666	242	146	71	5	0	0	0	0	0	3	46	153		
	HDD65	2637	642	496	376	141	29	0	0	0	3	101	326	523		
	CDD50	5455	67	74	174	376	620	827	937	906	732	458	182	102		
	CDD65	1951	2	4	13	63	184	377	472	441	285	92	12	6		
	CDH74	16285	7	25	104	487	1451	3309	4469	3843	1927	583	59	21		
	CDH80	5796	0	1	11	102	451	1298	1802	1458	559	111	3	0		
(14) Wind	WSAvg	6.6	6.9	7.5	8.0	8.1	7.0	6.2	5.8	5.3	6.1	5.8	6.0	6.2		
(15) Precipitation	PrecAvg	54.6	4.4	3.9	4.2	3.3	4.5	4.8	7.0	6.6	5.2	3.1	3.1	3.6		
	PrecMax	72.7	8.1	9.4	9.2	8.3	10.1	8.7	13.9	13.0	13.9	9.8	7.7	9.8		
	PrecMin	41.6	1.2	1.3	0.6	0.7	1.0	2.0	2.9	1.7	0.9	0.1	0.3	0.3		
	PrecStd	7.8	2.0	1.9	1.9	1.8	2.0	1.9	2.8	3.2	3.6	2.2	1.8	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.0	81.7	86.8	92.2	95.7	95.7	95.2	90.7	86.9	79.3	76.8		
		MCWB	64.8	65.6	65.0	69.0	73.6	77.1	79.5	78.9	75.9	73.5	69.7	67.8		
	2%	DB	70.4	72.2	77.0	82.9	88.2	92.3	93.3	92.1	88.1	83.1	75.4	72.6		
		MCWB	62.6	61.8	63.9	67.6	72.8	76.7	79.0	78.2	75.1	71.7	65.6	65.9		
	5%	DB	66.5	67.9	72.9	79.9	84.9	89.8	91.1	89.8	85.9	80.0	72.4	69.1		
		MCWB	60.3	60.0	61.6	66.3	71.3	76.0	78.1	77.6	74.0	69.8	64.9	64.1		
	10%	DB	62.4	64.0	69.0	76.2	81.9	87.3	88.8	87.5	83.4	77.1	69.1	65.4		
		MCWB	57.0	57.2	60.5	64.4	69.9	74.9	77.1	76.4	73.4	68.5	62.9	61.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	67.8	68.8	73.3	76.6	80.2	82.1	81.9	79.1	77.1	72.4	70.5		
		MCDB	71.3	74.4	75.0	80.4	86.9	90.3	91.7	90.5	86.1	81.7	76.2	74.4		
	2%	WB	64.7	65.2	66.5	70.6	74.8	78.5	80.5	80.2	77.7	74.9	69.2	67.5		
		MCDB	69.0	70.4	72.8	78.5	83.9	88.0	89.7	88.2	84.0	79.6	72.5	71.1		
	5%	WB	61.5	62.1	64.4	68.7	73.2	77.5	79.4	79.0	76.6	73.1	66.6	64.8		
		MCDB	66.0	66.2	70.4	76.1	81.4	86.7	88.0	86.5	82.3	77.5	70.5	68.1		
	10%	WB	57.8	58.1	62.1	66.8	71.7	76.4	78.3	78.0	75.5	70.9	63.9	61.8		
		MCDB	61.4	63.3	67.8	73.7	78.9	84.4	86.1	84.7	80.5	75.2	68.1	65.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	21.2	21.6	21.4	19.6	17.9	16.8	16.3	16.1	19.3	21.1	19.7		
		MCDBR	25.1	26.0	26.0	24.7	22.2	20.4	19.0	18.9	18.5	20.6	23.3	22.4		
		MCWBR	18.1	17.5	14.6	12.2	9.7	8.2	7.5	7.6	7.8	10.5	15.1	16.2		
	5% WB	MCDBR	22.5	23.9	21.3	20.8	19.0	18.4	17.5	17.0	15.4	15.7	19.5	20.1		
		MCWBR	18.7	18.6	14.7	12.1	8.9	8.0	7.7	7.5	7.3	9.2	15.0	16.6		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.330	0.357	0.399	0.444	0.493	0.522	0.507	0.420	0.369	0.337	0.324		
	taud		2.534	2.492	2.437	2.320	2.234	2.142	2.077	2.112	2.357	2.479	2.519	2.544		
	Ebn at Noon		283	291	290	281	269	254	246	247	266	273	272	272		
	Edh at Noon		26	30	34	40	44	49	52	49	37	30	26	24		
(44) All-Sky Solar Radiation	RadAvg		824	1054	1388	1756	1935	2017	1912	1736	1439	1205	912	731		
	RadStd		51	106	100	117	135	109	110	101	98	134	66	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+355	+236
Regional (45 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+207

Nomenclature: See separate page